

MATÍAS SOLORZA

Department of Economics
University of California, Davis

Email: masolorza@ucdavis.edu
Phone: +1 (530) 760-8438

Fields of Interest

Macroeconomics, Climate Economics

Education

University of California, Davis

2022 – Present

PhD in Economics

PhD Candidate, 2025 –

Universidad de Chile

2016

M.A. in Applied Economics (with distinction); Professional Degree in Industrial Engineering (highest distinction)

Research

Working Papers

- Briones, L., and Solorza, M. (2025). *Natural Disasters and Slow Recoveries: New Evidence from Chile*.

Central Bank of Chile Working Paper No. 1062.

Abstract: We study the macroeconomic responses of Chilean regions to natural disasters—floods and wildfires—using local projections and a combination of public and novel administrative data. We document persistent GDP losses, temporary declines in consumption, and a delayed recovery in investment, accompanied by rising employment but falling wages and effective hours. These dynamics contrast with U.S. county-level evidence and point to the importance of disaster size, as well as institutional and financial conditions, in shaping post-disaster recoveries in emerging economies. We interpret the evidence through four mechanisms: destruction of productive capital, tighter financial conditions that constrain rebuilding, reallocation of production, and household wealth losses that depress consumption while supporting low-wage reconstruction employment. Embedding these mechanisms into a real business cycle model with financial frictions, we show that financial constraints are quantitatively central: absent these frictions, post-disaster losses in economic activity during the first years would be reduced by about one half. Our findings highlight the role of targeted financial and reconstruction policies in mitigating the long-run economic costs of climate-related disasters.

Work in progress

- The Ambiguity Costs of Climate Change (with Frances Moore and Benjamin Collier)

Abstract: Weather risks are fundamentally more difficult to manage under climate change because past experience of weather is potentially unrepresentative of current exposure. The possibility of climate change acts as a global information shock, degrading utility of the historic weather record with implications for actors across the economy making weather-contingent decisions. Here

we quantify these ambiguity-related costs for a major climate risk, namely urban flood damages in the United States. Using daily rainfall for 151 US cities from 1970 to 2024, we contrast the 2024 extreme rainfall distribution under a Bayesian model that assumes the rainfall distribution is stationary with one that allows for time-evolving GEV parameters, consistent with the possibility of a changing climate. Estimated tail risks increase notably when allowing the climate to evolve: the 99th percentile rainfall event is larger in 89% of the cities in our sample. Economic costs of these tail risks are amplified by convex damage functions, which we estimate at the city level using flood insurance policy and claim data. Modeling aggregate losses for portfolios at the city, state, regional, and national level, we find geographic diversification only modestly abates these tail risks: the 99th percentile loss event is 4-5 times larger when accounting for evolving climate risk at all levels of aggregation. In a private insurance market these much larger tail risks would substantially increase required capital holdings and the cost of supplying insurance. Our analysis highlights the substantial economic costs of climate change-driven ambiguity in weather extremes and sheds light on recent volatility in US insurance markets.

- How firms' investment decisions change when they are exposed to natural disasters

Publications

- Arroyo Marioli, F., Becerra, J. S., & Solorza, M. (2022). *The credit channel in Chile through the lens of a semi-structural model*. *Latin American Journal of Central Banking*, 3(2).

Professional Experience

Central Bank of Chile — Medium-Term Forecasting Department

- Senior Economic Analyst, 2021–2022
- Economic Analyst, 2019–2021

Banchile Inversiones — Research Department

Economic Analyst, 2017–2019

Policy Contributions and Notes

- “Impacts of Macroeconomic Policies during the COVID-19 Crisis.” *Monetary Policy Report*, September 2021, Central Bank of Chile (Spanish).
- “Estimation of Structural Parameters for the Chilean Economy.” *Monetary Policy Report*, June 2021, Central Bank of Chile (Spanish).
- “Use of Macroeconomic Models in the Central Bank of Chile.” Central Bank of Chile, December 2020 (Spanish).

(*) Joint work with multiple colleagues at the Central Bank of Chile.

Teaching Assistant Experience (selected)

- Time Series Analysis (PhD), University of California, Davis.
- Macroeconomic Modelling for Emerging Economies (MSc/PhD), Universidad de Chile (2021).
- Introduction to Economics; Macroeconomics; Industrial Organization, Universidad de Chile (2014–2016).

Seminars and Conference Presentations

- Modeling Climate-Economic Dynamics Workshop for Young Scholars, University of Chicago (2026, forthcoming); 1st Piraeus International Economic Conference, Spetses, Greece (2026, forthcoming); World Congress of Environmental and Resource Economists (WCERE), Carcavelos, Portugal (2026, forthcoming); International Conference on Computing in Economics and Finance (2025); Annual Meeting of the Chilean Economic Society (SECHI), Chile (2025); Universidad de Talca, Chile (2025).
- Internal seminars: Central Bank of Chile (2024); UC Davis (2025); NatuRE Policy Lab (2025).
- Pre-PhD: Annual Meeting of the Chilean Economic Society (SECHI), Chile (2016).

Grants, Awards, and Honors

- Dean's Summer Graduate Support Award, University of California, Davis, 2026.
- Selected for the Internship Program for Ph.D. Students, Central Bank of Chile, 2024–2025.
- Travel Award for Conference Participation in Europe, University of California, Davis, 2026.
- Travel Award, “The Macroeconomics of Climate Change” Summer School, CEMFI, 2023.
- PhD Fellowship, Central Bank of Chile, 2022.
- Funding for Master's Thesis, CONICYT (Chile), 2015.
- Scholarship for Master's Program, Center for Applied Economics (CEA), Universidad de Chile, 2015.
- Outstanding Student Award, Universidad de Chile, 2013–2014.

Computational Skills

Matlab; R; Julia; Stata; EViews; $\text{\LaTeX}$; Bloomberg

Languages

English (fluent); Spanish (native)

References

Òscar Jordà

Department of Economics
University of California, Davis
ojorda@ucdavis.edu

Frances Moore

Department of Environmental Science and Policy
University of California, Davis
fmoore@ucdavis.edu

Nicolás Caramp

Department of Economics
University of California, Davis
ncaramp@ucdavis.edu

Emile Marin

Department of Economics
University of California, Davis
emarin@ucdavis.edu

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